If $P(A/B) = P(A)$, then

$$\frac{P(A \cap B)}{P(B)} = P(A).$$

$$\Rightarrow \quad P(A \cap B) = P(A) \times P(B).$$

Similarly, the relation $P(B/A) = P(B)$ also indicates the independence of the events A and B.

Definition :

Two events A and B are said to be independent of one another, if

$$P(A \cap B) = P(A) \times P(B).$$



NOTE

The greater the value of β_2, the more peaked the distribution.

(i) If A and B are not independent events, they are called dependent events.

(ii) The above definition of independence may be extended to finite number of events. If the events $A_1, A_2, \ldots, A_n$ satisfy

$$P(A_1 \cap A_2 \cap \ldots\ldots A_n) = P(A_1) \times P(A_2) \times \ldots \times P(A_n),$$

then $A_1, A_2, \ldots, A_n$ are called independent events.

Example 8.16

In tossing a fair coin twice, let the events A and B be defined as A: getting head on the first toss, B: getting head on the second toss. Prove that A and B are independent events.

Solution:

The sample space of this experiment is

$$S = \{HH, HT, TH, TT\}.$$

The unconditional probabilities of A and B are $P(A) = \frac{1}{2} = P(B)$.

The event of getting heads in both the tosses is represented by $A \cap B$. The outcome of the experiment in favour of the occurrence of this event is HH. Hence, $P(A \cap B) = \frac{1}{4}$.

$\therefore$ $P(A \cap B) = P(A) \times P(B)$ holds. Thus, the events A and B are independent events.